

Lecture 1, Dec 17, 2025. Inequalities and Absolute Value.

In the real numbers we have two possibilities for two arbitrary numbers.

i). The numbers are equal " $=$ "

for example: $1, \frac{5}{5}$ or $2, \sqrt[3]{8}$

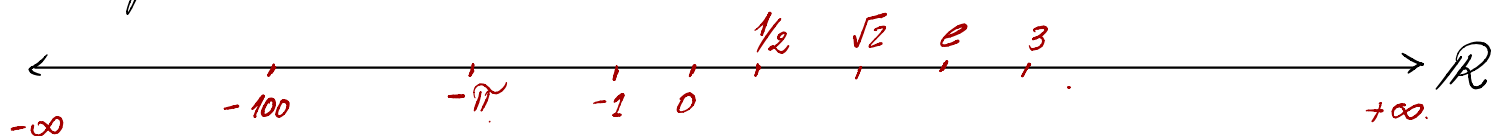
ii). The numbers are different, but in this case we can ask yourself about which is greater than the another or less than the other.

In general if x, y are real numbers then:

$$x = y \text{ or } x < y \text{ or } x > y.$$

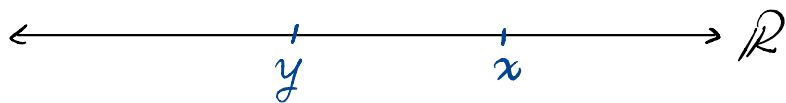
" x equal to y " " x less than y " " x greater than y "

Using this property we can draw all the real numbers in a number line representation.

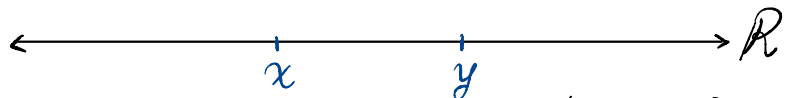


In this line there are all the real numbers and we can give an interpretation for $x > y$ or $x < y$ geometrically.

We say $x > y$ if x is in the right side of y .

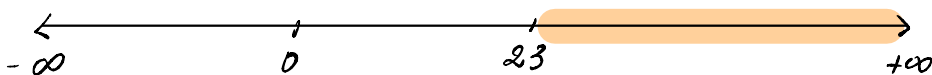


We say $x < y$ if x is in the left side of y .



The symbols $x \geq y$ and $z \leq y$ are true if $x = y$ or $x > y$ and $z = y$ or $z < y$

If you fix z , then which are the numbers greater than z ?



we have infinite numbers.

Def. (Set) A set is a collection (sometimes infinite) of objects, and the objects are called the elements.

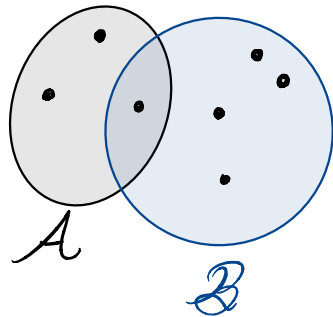
Notation: • If S is a set the notation $a \in S$ means a is an element of S , otherwise $a \notin S$.

For example: $N = \text{natural numbers} = \{0, 1, 2, 3, 4, 5, \dots\}$.

Notice that $5 \in N$ but $\sqrt{2} \notin N$.

Operations Between Sets

i). Union: If A and B are two sets then the union $A \cup B$ is the set consisting of elements that are in A or B .



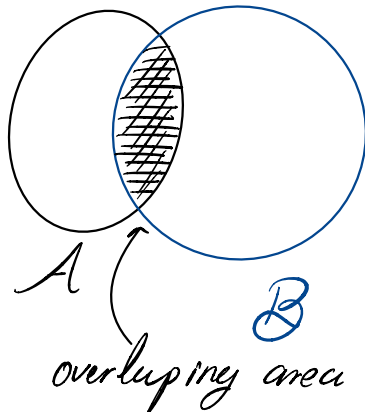
$A \cup B = \text{everything}$.

Example: $A = \{1, 2, \square\}$ $B = \{\star, \frac{1}{2}, -1\}$.

then

$$A \cup B = \{1, 2, \square, \star, \frac{1}{2}, -1\}$$

ii). Intersection: If A and B are two sets then $A \cap B$ is the set consisting of elements in A and B .



Example: $A = \{\star, 1, \pi, e\}$ $B = \{7, 32, -15, \star\}$.

$$A \cap B = \{\star\}$$

If A and B don't have common elements then $A \cap B = \underline{\Phi}$.

Φ is the set that contains no elements

More examples of sets:

$A = \{1, 2, 3, 4, 5, 6\}$ in other notation

$$A = \{x \mid x \text{ is natural number and } 0 < x < 7\}$$

A particular case of sets:

Intervals:

Geometrically

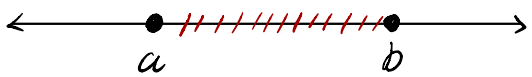
Set Notation

Open interval:



$$(a,b) = \{x \in \mathbb{R} \mid a < x < b\}$$

Closed interval.



$$[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$$

Combinations of these intervals.

$$(a, b] = \text{---} \underset{a}{\circ} \text{ // // // // // // // } \underset{b}{\bullet} \text{ ---} = \{x \in \mathbb{R} \mid a < x \leq b\}.$$

$$[a, b) = \text{---} \bullet \text{-----} \circ \text{---} = \{x \in \mathbb{R} \mid a \leq x < b\}.$$

$$(a, \infty) = \text{---} \underset{a}{\circ} \text{+++++} \rightarrow = \{x \in \mathbb{R} \mid x > a\}.$$

$$[a, \infty) = \text{---} \bullet_a \text{-----} \rightarrow = \{x \in \mathbb{R} \mid x \geq a\}.$$

$$(-\infty, b) = \begin{array}{c} a \\ \leftarrow \text{-----} \bigcirc \text{-----} \\ b \end{array} = \{x \in \mathbb{R} \mid x < b\}$$

$$(-\infty, b] = \leftarrow \text{|||||} \underset{\substack{b \\ b}}{\bullet} = \{x \in \mathbb{R} \mid x \leq b\}$$

$$(-\infty, \infty) = \langle \text{+++++} \rangle = \mathbb{R}.$$

Problem: A bus can carry a maximum of 65 passengers or less. If on tip, the bus already carries 30 people, how many more people can get on the bus?

Model: Let x the number of additional people. So,
 $30 + x \leq 65$ (an inequality).

Goal: We want describe x .

Rules for inequalities:

- i). $a < b$ then $a + c < b + c$.
- ii). if $a < b$ and $c < d$ then $a + c < b + d$.
- iii). if $a < b$ and $c > 0$ then $ac < bc$.
- iv). if $a < b$ and $c < 0$ then $ac > bc$.
- v). if $0 < a < b$ then $\frac{1}{b} < \frac{1}{a}$.

with these properties we can solve inequalities.

From i). $c = -30$ then $x + 30 \leq 65$.

$$x + 30 + (-30) \leq 65 + (-30)$$

$$x \leq 65 - 30$$

$$x \leq 35.$$

So, Ans: 35 or fewer people can be on the bus.

Example: Solve $1 + x < 7x + 5$

Sln: Step 1: All the variables in one side (left or right).

Hint: try keep the coefficient of the variable positive.

$$1 < 7x + 5 - x$$

$$1 - 5 < 7x - x$$

$$-4 < 6x$$

$$\frac{-4}{6} < x$$

$$-\frac{2}{3} < x \quad \text{So}$$

$$\leftarrow \text{---} \frac{-2}{3} \text{---} \rightarrow \quad \mathcal{S} = \left(-\frac{2}{3}, \infty\right)$$

Solve the inequality: $x^2 - 5x + 6 \leq 0$.

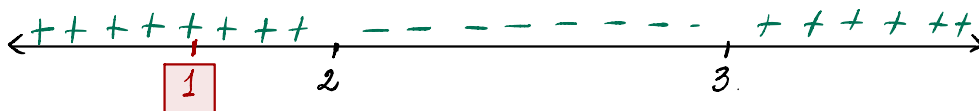
Step 1 Unequal to zero (this step is free here)

Step 2 Factorize

$$x^2 - 5x + 6 \leq 0$$

$$(x - 3)(x - 2) \leq 0$$

Step 3 Draw a line with the roots of the polynomials above.



Step 4 Select "easy" number and evaluate the sign.

$$\text{if } x=1, \quad (1-3)(1-2) = (-2)(-1) = 2 > 0 \quad (\text{positive})$$

Step 5 Select the region for your problem.

We are looking for the negative numbers.

$$(x-3)(x-2) \leq 0 \quad (\text{negative numbers})$$

$$\mathcal{S} = [2, 3]$$

For you: Solve $x^3 + 3x^2 > 4x$.

Absolute Value the absolute value of a number a , denoted by $|a|$ is the distance from a to 0 on the real line.

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases}$$

Problem. Solve

$$\underbrace{|3x-2|}_a = \begin{cases} \underbrace{3x-2}_a & \text{if } \underbrace{3x-2 \geq 0}_a \\ \underbrace{-(3x-2)}_{-a} & \text{if } \underbrace{3x-2 < 0}_a \end{cases}$$

Solving

$$3x-2 \geq 0$$

$$3x-2 < 0$$

$$3x \geq 2$$

and

$$3x < 2$$

$$x \geq \frac{2}{3}$$

$$x < \frac{2}{3}$$

$$|3x-2| = \begin{cases} 3x-2 & \text{if } x \geq \frac{2}{3} \\ 2-3x & \text{if } x < \frac{2}{3} \end{cases}$$

Properties: i). $|a \cdot b| = |a| \cdot |b|$

$$\text{ii). } \left| \frac{a}{b} \right| = \frac{|a|}{|b|} \quad \text{if } b \neq 0.$$

$$\text{iii). } |a^n| = |a|^n$$

Now assume $a > 0$ then.

$$\text{i). } |x| = a \quad \text{if and only if } x = \pm a.$$

$$\text{ii). } |x| < a \quad \text{if and only if } -a < x < a. \quad (-a < x \text{ and } x < a)$$

$$\text{iii). } |x| > a \quad \text{if and only if } x > a \text{ or } x < -a$$

Example: i). $|2x-5|=3 \stackrel{\text{def}}{\Leftrightarrow} 2x-5=3 \quad \text{or} \quad 2x-5=-3$

$$2x = 3+5$$

$$2x = -3+5$$

$$2x = 8$$

$$2x = 2$$

$$x = \frac{8}{2}$$

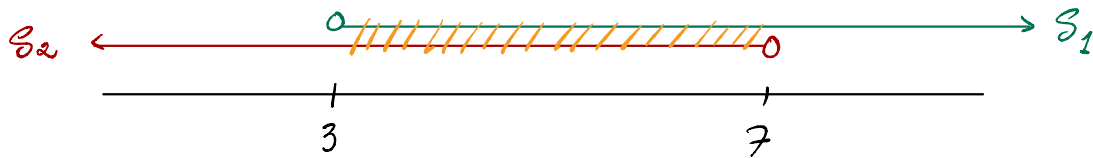
$$x = \frac{2}{2}$$

$$x = 4$$

$$x = 1$$

$$\begin{array}{lll}
 \text{ii). } |x-5| < 2 & \stackrel{\text{def}}{\Leftrightarrow} & -2 < x-5 \quad \text{and} \quad x-5 < 2. \\
 & & -2+5 < x & x < 2+5 \\
 & & 3 < x & x < 7 \\
 & & S_1 = (3, \infty) & \text{intersection } S_2 = (-\infty, 7)
 \end{array}$$

$$S = S_1 \cap S_2$$



$$S = (3, 7)$$